

QUOC VAN NGUYEN

Madison, WI • (608) 327-9872 • qvnguyen3@wisc.edu

quoc-nguyen-portfolio-six.vercel.app • linkedin.com/in/quoc-nguyenvan • github.com/koshiroquoc

EDUCATION

University of Wisconsin–Madison

Expected May 2028

B.S. in Data Science, Statistics, and Mathematics | GPA: 3.80 / 4.00

Madison, WI

- Mathematics for Data Science option · Certificate in Economic Analytics · Dean's List: Fall 2024, Spring 2025, Fall 2025
- **Coursework:** Matrix Methods in Machine Learning, Machine Learning & Statistical Pattern Classification, Data Management for Data Science (SQL, ETL, warehousing), Data Science Programming I–II, Probability Theory, Linear Algebra, Linear Optimization, Intermediate Microeconomics. *In progress:* Applied Econometrics, Data Science Modeling II, Theory of Single Variable Calculus.

PROJECTS

Divvy Bikeshare Demand Forecasting | Python, LightGBM, Polars, MLflow, FastAPI, Prefect, Docker

Summer 2026

- Built an end-to-end system forecasting station-hour trip demand 24 hours ahead for the top 200 Chicago Divvy stations — 65% of annual trip volume — to support bicycle rebalancing decisions.
- Trained LightGBM with a Poisson objective on leak-safe lag and rolling features, reaching mean MAE 1.101 ± 0.283 across four rolling-origin monthly backtest windows — **24.6% below a seasonal naive baseline** and 8.1% below historical mean.
- Shipped production-shaped infrastructure behind **51 automated tests** covering leakage guards and train/serve consistency: MLflow tracking, champion/challenger promotion gates, Evidently drift monitoring, and containerized CI delivery via GitHub Actions.

Protected Bike Lanes and Divvy Ridership — Causal Study | Python, staggered DiD, panel data

July 2026

- Designed a staggered difference-in-differences study of 12 newly completed protected bike-lane corridors, comparing 40 treated stations against 83 matched control stations across a 36-month station-month panel.
- Estimated a matched group-time ATT with two-way clustered inference, locking the design, thresholds, and estimator roles **before reading any effect**; validated with donut/radius variants, leave-one-corridor-out, and two placebo tests.
- Reported a -10.8% point estimate (95% CI $[-24.0\%, +2.3\%]$) and a failed geography placebo, retaining both in the final report rather than tuning them away; `make reproduce` rebuilds the analysis under 28 automated tests.

Da Nang Tour Guide — Evaluation-Driven RAG System | Elasticsearch, Gemini API, Streamlit, Docker

2025 – 2026

- Built a hybrid retrieval pipeline fusing BM25 and dense vector search via Reciprocal Rank Fusion over a reproducible 2,557-place OpenStreetMap snapshot with per-record provenance, licensing, and SHA-256 build metadata.
- Raised discovery **MRR@5 to 0.693** (+0.269 over BM25, paired-bootstrap 95% CI $[+0.200, +0.339]$) on a 210-query retrieval benchmark; a Gemini self-query planner guarded by data-derived allowlists and a deterministic validator reached **MRR@5 0.957** (+0.264, 95% CI $[+0.196, +0.334]$) on a separate 120-query benchmark at 97.5% exact filter match.
- Found that $\sim 80\%$ of the original benchmark questions leaked the venue name, then split the benchmark into 120 discovery and 90 name-lookup queries and re-ran every comparison on honest metrics.

EXPERIENCE

AI Engineer Intern

May 2025 – Aug 2025

DataHouse Asia

Da Nang, Vietnam

- Contributed to an internal HR chatbot built on retrieval-augmented generation, focusing on reducing hallucination in generated responses.
- Iterated on prompt design and retrieval configuration, evaluating output quality against source documents across successive versions.

Shift Lead

Oct 2024 – Jul 2026

University of Wisconsin–Madison, Division of Housing

Madison, WI

- Promoted to Shift Lead within five months; led approximately 10 student workers per shift and coordinated daily operations with full-time staff.
- Trained and mentored 50+ new employees on operational workflows, safety protocols, and cross-team collaboration.

LEADERSHIP & INVOLVEMENT

Cultural Chair, Vietnamese International Student Association — UW–Madison

2024 – Present

Manage member communications and organize cultural and community events for the Vietnamese student community.

SAT Tutor Volunteer, AimHigh — UW–Madison

Jan – May 2025

Delivered 100+ hours of pro-bono SAT tutoring with individualized lesson plans for standardized-test readiness.

TECHNICAL SKILLS

Languages: Python, R, SQL

Machine Learning & Statistics: scikit-learn, LightGBM, XGBoost, CatBoost, time-series forecasting, difference-in-differences, group-time ATT, bootstrap confidence intervals, LASSO, cross-validation, placebo & falsification testing

Data & Infrastructure: Polars, Parquet, DuckDB, Kafka, Prefect, Elasticsearch, FastAPI, Docker & Compose, GitHub Actions, MLflow, Evidently, Grafana, pytest, Streamlit